

PHASE EQUILIBRIA, STRUCTURE, AND PROPERTIES OF COMPLEX OXIDES IN THE $\text{Eu}_{1-x}\text{Sr}_x\text{Co}_{1-y}\text{Fe}_y\text{O}_{3-\delta}$ SYSTEM AT 1373 K IN AIR

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The aim of this work is a systematic study of phase equilibria, determination of the homogeneity ranges, and investigation of the structural features of complex oxides in the quasi-quaternary system $\text{EuCoO}_3 - \text{SrCoO}_{3-\delta} - \text{SrFeO}_{3-\delta} - \text{EuFeO}_3$.

A series of samples with the composition $\text{Eu}_{1-x}\text{Sr}_x\text{Co}_{1-y}\text{Fe}_y\text{O}_{3-\delta}$ with $0.1 \leq x \leq 0.9$ and $0.1 \leq y \leq 0.9$ was synthesized by a glycerol-nitrate technique, followed by final annealing at 1373 K in air and subsequent quenching to room temperature. The phase composition and crystal structure were studied by X-ray diffraction and the Rietveld full-profile analysis.

An isobaric-isothermal section of the phase diagram of the quasi-quaternary system $\text{EuCoO}_3 - \text{SrCoO}_{3-\delta} - \text{SrFeO}_{3-\delta} - \text{EuFeO}_3$ at 1373 K in air has been constructed. It was found that depending on the strontium (x) and iron (y) content the solid solutions $\text{Eu}_{1-x}\text{Sr}_x\text{Co}_{1-y}\text{Fe}_y\text{O}_{3-\delta}$ crystallize in one of three perovskite structures: cubic (sp. gr. $Pm\bar{3}m$), tetragonal (sp. gr. $I4/mmm$), or orthorhombic (sp. gr. $Pbnm$). The cubic phase is stable for compositions enriched in strontium and iron ($x = 0.9$ and $0.1 \leq y \leq 0.9$; $x = 0.8$ and $0.2 \leq y \leq 0.9$; $x = 0.7$ and $0.5 \leq y \leq 0.9$). The tetragonal superstructure $2a_p \times 2a_p \times 4a_p$, characterized by an ordered arrangement of Eu and Sr cations in the A-sublattice, observed for $x = 0.8$ and $y = 0.1$; for $x = 0.7$ and $0.1 \leq y \leq 0.4$; for $x = 0.6$ and $0.1 \leq y \leq 0.3$, while the orthorhombic structure is characteristic of europium-rich oxides with $x = 0.1$ and $0.1 \leq y \leq 0.9$. The unit cell parameters exhibit a linear dependence on composition upon substitution of europium by strontium in the A-site and of cobalt by iron in the B-site obeying Vegard's rule.

Thermogravimetric analysis of the oxides $\text{Eu}_{1-x}\text{Sr}_x\text{Co}_{1-y}\text{Fe}_y\text{O}_{3-\delta}$ ($0.6 \leq x \leq 0.9$ and $y = 0.3$; $x = 0.8$ and $0.3 \leq y \leq 0.8$) showed that all of them are oxygen-deficient. The oxygen content, determined by iodometric titration and reduction in a hydrogen stream, decreases with increasing temperature and increasing strontium and cobalt contents in the samples. Charge compensation upon substitution of Eu^{3+} by Sr^{2+} occurs through the formation of oxygen vacancies and an increase in the average oxidation state of 3d-metals.

The nonlinear temperature dependence of relative thermal expansion of $\text{Eu}_{1-x}\text{Sr}_x\text{Co}_{0.7}\text{Fe}_{0.3}\text{O}_{3-\delta}$ ceramic samples ($x = 0.6, 0.7$) in the range of 300–1373 K in air is caused by thermal and chemical factors. The latter is associated with the release of oxygen from the lattice at $T > 600$ K and a change in the oxidation state of 3d-metals. The increase in LTEC with increasing strontium content correlates with the increase in the oxygen nonstoichiometry.

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